LPS/C/34

II PERIODIC TEST : 2025-26

CLASS – X (CBSE)

SCIENCE [086]

Maximum Marks : 50
Time allowed : Two hours

General Instructions : -

- This question paper comprises of four Sections - A, B, C and D.
- All questions are compulsory.
- Section - A consists of 1 mark questions, it contains MCQ & case study type questions.
- Section - B consists of 1 mark questions, it contains very short answer and Assertion - Reason type questions.
- Section - C consists of 2 marks questions, it contains short answer type questions.
- Section - D consists of 5 marks questions, it contains long answer type questions.

SECTION-A [17 Marks]

[BIOLOGY]

Q.1. What prevents backflow of blood inside the heart during contraction? [1]

(a) valves in heart (b) Thick muscular walls of ventricles
(c) Thin walls of atria (d) All of the above

Q.2. One of the events that does not occur during photosynthesis: [1]

(a) Chlorophyll absorbs solar energy
(b) Carbon dioxide is released during the process
(c) Oxygen is released during the process
(d) Carbon dioxide is absorbed during the process

Q.3. Which one among the following is not removed as a waste product from the body of a plant? [1]

(a) Resins and gums (b) Urea
(c) Dry leaves (d) Excess water

Q.4. **Read the following and answer the questions that follow :** **[1×3=3]**

Arteries, veins and capillaries are blood vessels through which blood flows in our body. Arteries carry blood from heart to different parts of the body whereas viens deliver blood back to the heart. Arteries are connected to vein by thin capillaries.

(i) Which two chambers of the human heart receive oxygenated and deoxygenated blood from different parts of the body?
(ii) Which blood vessels carry deoxygenated blood from the heart to the lungs?
(iii) Walls of ventricle is much thicker than the wall of atria. Why?

Q.5. (a) Write the name of enzymes present in gastric juice. [1]
(b) Write the name of chemical substance that is produced during anaerobic respiration in human muscles. [1]
(c) Write the name of blood vessel that carries blood to kidney for filtration. [1]

Q.6. **It consist of two statements Assertion and Reason - choose the correct option which are given below:**

Assertion : In wood plants gases exchange occur through lenticels. [1]
Reason : Lenticels are specialized cells found along the stomata in woody plants.

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true but Reason is the correct explanation of Assertion.
(c) Only Assertion is true but Reason is false.
(d) Assertin is false but reason is true.

Q. 7. Give reason : [2]

(a) Small intestine in herbivores is longer than carnivores.
(b) Wind pipe is supported by 'C' shaped cartilaginous ring.

Q. 8. (a) Name the hormones secreted by the following endocrine glands and specify one function of each. [3]

(i) Thyroid (ii) Pituitary (iii) pancrease

(b) We are advised to take iodised salt in our diet by doctors. Justify its importance in our body. [2]

OR

(a) State the structural and functional unit of the nervous system. Draw its neat labelled diagram. [3]

(b) Name the parts of human brain which controls : [2]

(i) Muscular coordination and body balance.

(ii) Seat of intelligence

SECTION-B [16 Marks]
[CHEMISTRY]

Q.9. Which of the following statement is true for an amphoteric oxide? [1]

(a) It reacts only with acid and does not form water.

(b) It reacts acid as wall as base to form salt and hydrogen gas

(c) It reacts with both acid an well as base to form salt and water

(d) It rects only with base and does not form water

Q.10. Electrolysis of water is a decomposition reaction where O_2 and H_2 are released. Find the ratio of the volume of hydrogen and oxygen released at electrodes. [1]

(a) 1 : 1 (b) 2 : 1 (c) 4 : 1 (d) 1 : 2

Q.11. What is the chemical formula and chemical name of Plaster of Paris? [1]

(a) $CaSO_4.2H_2O$, calcium sulphate dihydrate

(b) $CaSO_4 \cdot \frac{1}{2}H_2O$, calcium sulphate hydrate

(c) $CaSO_4.H_2O$, calcium sulphate semi hydrate

(d) $CaSO_4 \cdot \frac{1}{2}H_2O$ calcium sulphate hemi hydrate

Q.12. **Read the passage carefully and answer the questions that follows:** [4]

Ritika conducted a series of experiments in which she used different chemical reactions while working upon a project related to chemistry. The given experiments which she performed are :

Experiment A : She mixed $BaCl_2$ (Barium chloride) with Na_2SO_4 solution. In this experiment a white ppt was obtained.

Experiment B : She burned magnesium ribbon in air where she observed that it burns with dazzling flame.

Experiment C : In this experiment she kept lime stone in a china dish and heated it strongly.

Now she needs your help to solve these questions:

(i) Write the name of type of reaction used in experiment 'A'.

(ii) Write the balance chemcial reaction taken place in experiment 'A'.

(iii) Write the name and colour of the substance formed after burning magnesium in air in experiment 'B'.

(iv) Write the names of the products obtained in experiment 'C'.

Q.13. What will happen if a solution of sodium hydrogen carbonate its heated? Give equation of the reaction. [1]

Q.14. Write reaction of dry slaked lime ($Ca(OH)_2$ with chlorine. [1]

Q.15. **It consist of two statements Assertion and Reason - choose the correct option which are given below:**

Assertion : Gallium and caesium will get melt on your palm if you keep these metals on your palm. [1]

Reason : Melting point of these metals is low.

(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(b) Both Assertion and Reason are true but Reason is the correct explanation of Assertion.

(c) Only Assertion is true but Reason is false.

(d) Assertin is false but reason is true.

Q.16. Compound 'x' and aluminium are used to join railway tracks. [2]
(i) Identify the compound 'x'.
(ii) Name the reaction.
(iii) Write down equation of this reaction.

OR

Show the formation of Na_2O and MgO by the transfer of electrons.

Q.17. (a) A chemical compound 'X' is used in glass and soap industry. Identify the compound and give its chemical formula. [2]
(b) How many molecules of water of crystallisation are present in the compound 'x'.

Q.18. How is copper extracted from its sulphide ore? Explain the various steps supported by chemical equations. [2]

SECTION – C [17 Marks]

Q.19. A full length image of a distant tall building can definitely be seen by using: [1]
(a) A concave mirror (b) A convex mirror
(c) A plane mirror (d) Both concave and plane mirror

Q.20. If a beam of red light and a beam of violet light are incident at the same angle on the inclined surface of a prism from air and produce angles of refraction r and v respectively. Which of the following is correct ? [1]

(a) $r = v$ (b) $r > v$ (c) $r = \frac{1}{v}$ (d) $r < v$

Q.21. In torches, search lights and headlights of vehicles the bulb is placed : [1]
(a) between the pole and the focus of the reflector
(b) very near to the focus of the reflector
(c) between the focus and centre of curvature of the reflector
(d) at the centre of the curvature of the reflector

Q.22. Electrical resistivity of a given metallic wire depends upon : [1]
(a) Its length (b) Its thickness
(c) its shape (d) Nature of the material

Q.23. **Read the following and answer the questions that follow.** [1×3=3]
Absolute refractive index of some of the materials A, B, C and D are given in the following table:

Medium	Refractive index
A	1.54
B	1.33
C	2.42
D	1.65

(i) What is the meaning of the statement that 'Absolute refractive index of water is 1.33'?
(ii) How is absolute refractive index related to speed of light?
(iii) If speed of light in medium C is 'X' ms^{-1} then what will be speed of light in medium B in terms of 'X' ?

Q.24. A person wants to measure the potential difference across a conductor. Which device will be use? [1]

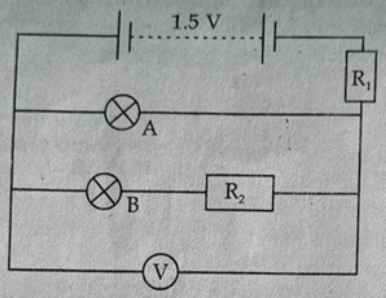
Q.25. Name the part of an eye which is equivalent to : [1]
(i) Diaphragm
(ii) Photographic plate in camera

Q.26. **In this question there are two statements Assertion and Reason. Choose the correct option which are given below:** [1]
Assertion(A): Electric current will not flow between two charged bodies when connected, if their charges are same.
Reason (R) : Current in the rate of flow of charge.
(a) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(b) Both Assertion and Reason are true but Reason is the correct explanation of Assertion.
(c) Only Assertion is true but Reason is false.
(d) Assertin is false but reason is true.

Q.27. A person uses a spectacle of focal length - 50 cm. [2]
(a) Name the defect of vision he is suffering from.
(b) List two main causes of developing this defect.

Q.28. A student has three concave mirrors A, B, and C of focal length 20 cm, 15 cm and 10 cm respectively. For each concave mirror, he performs the experiments of image formation for three values of object distance of 30 cm, 10 cm and 20 cm.
Give reason for the following:
(a) For the three object distances, identify the mirror which will form an image equal in size to that of object. Find at least one value of object distance. [1]
(b) Out of the three mirrors, identify the mirror which would be preffered to be used for shaving purpose? [1]
(c) Find the nature, position and size of image formed by mirror A for the object distance equal to 10 cm. [2]
(d) Draw the ray diagram for the case asked in part C. [1]

OR



As shown in the figure above A and B are two lamps. Lamp 'A' is rated at 12 V, Lamp B is rated at 6V. When lamp B operates at its rated voltage, the current in it is 3A.
When lamp A operates at its rated voltage the current in it is 2A.
The value of R_1 and R_2 are chosen so that both of lamps operates at their rated voltage.
Based on information given, answer the following :
(a) Calculate the resistance of lamp A. [1]
(b) State and give reason for the reading of the voltmeter. [1]
(c) Calculate the Resistance of R_2. [1]
(d) Find the value of the resistance R_1. [2]

Q1. (a) valves in heart
Q2. (b) CO_2 is released during the process
Q3. (b) Urea

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Q4. (i) Left atrium ($-O_2$ blood)
Right atrium (Deoxy
(ii) Pulmonary artery
(iii) Ventricles have to pump blood into various organs.

Q5. (a) Pepsin
(b) Lactic Acid
(c) Renal artery

Q6. (c)